

LAPORAN PRAKTIKUM
TEKNOLOGI IOT II PRAKTIKUM 1:
Arsitektur Internet of Things dan Mikrokontroler

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Disusun Oleh :

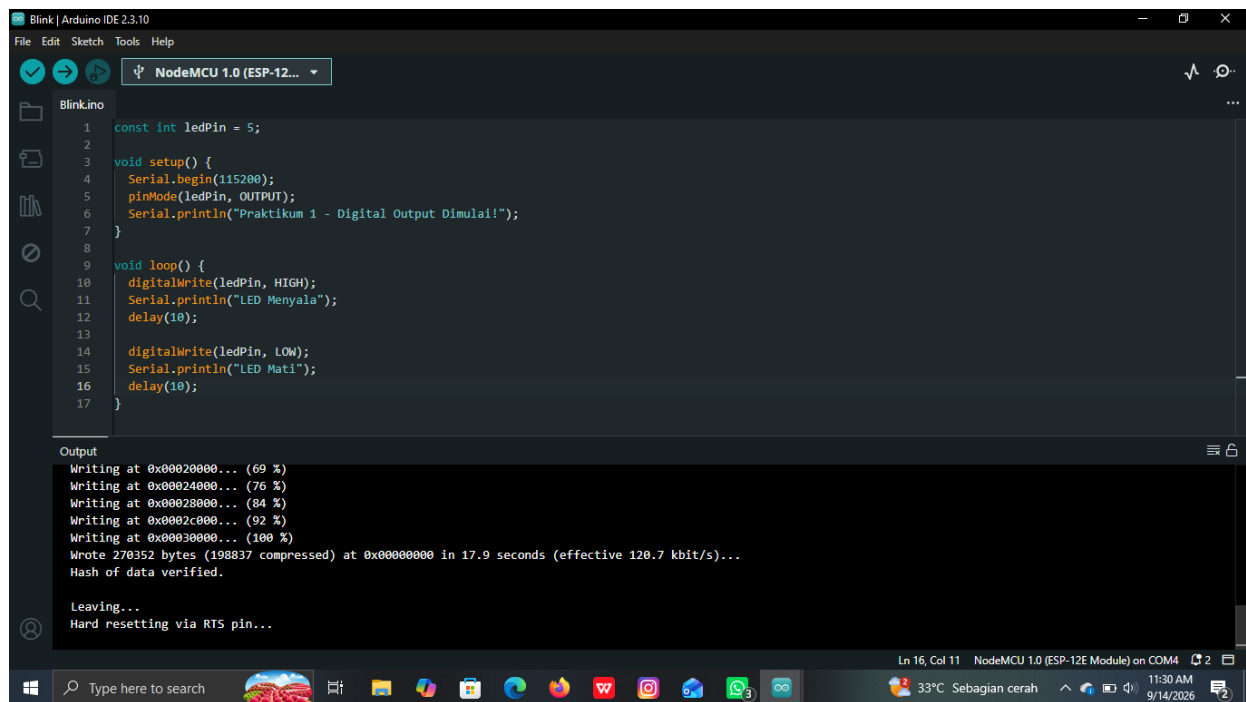
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PROGRAM STUDI TEKNOLOGI REKAYASA INTERNET
JURUSAN TEKNOLOGI INFORMASI
POLITEKNIK NEGERI LAMPUNG
TAHUN 2026

1.5.1 Praktikum 1: Digital Output (Menyalakan LED)



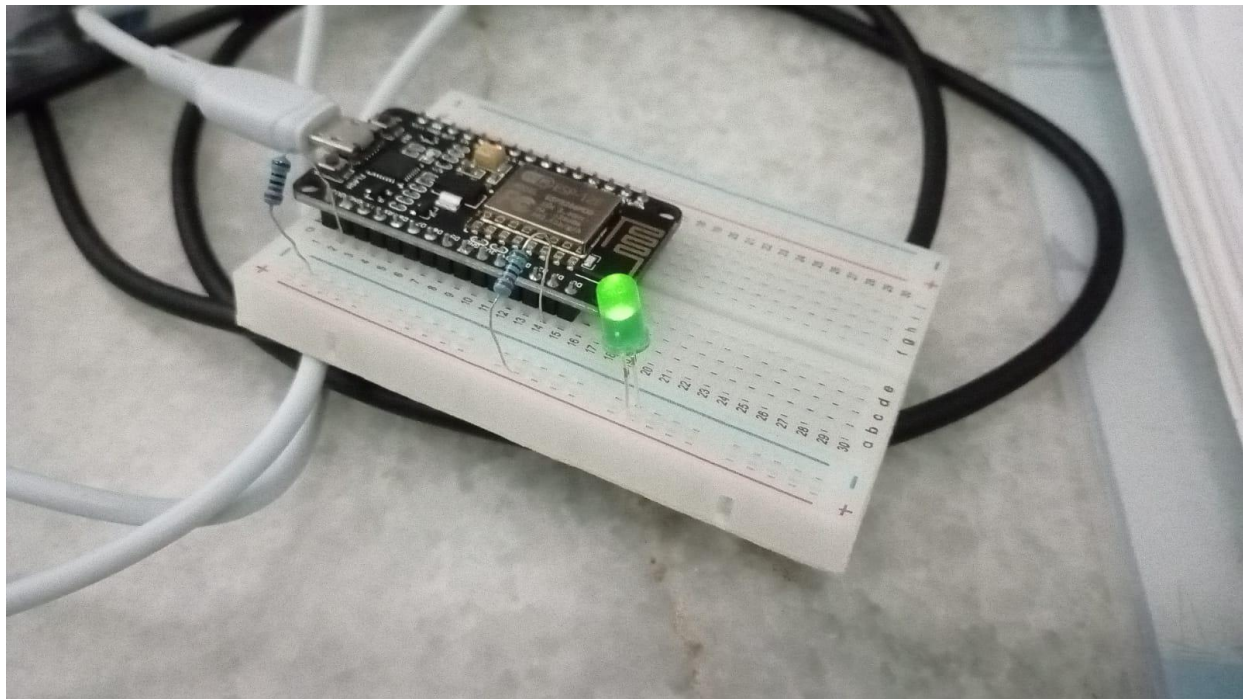
```
1 const int ledPin = 5;
2
3 void setup() {
4   Serial.begin(115200);
5   pinMode(ledPin, OUTPUT);
6   Serial.println("Praktikum 1 - Digital Output Dimulai!");
7 }
8
9 void loop() {
10  digitalWrite(ledPin, HIGH);
11  Serial.println("LED Menyala");
12  delay(10);
13
14  digitalWrite(ledPin, LOW);
15  Serial.println("LED Mati");
16  delay(10);
17 }
```

Output

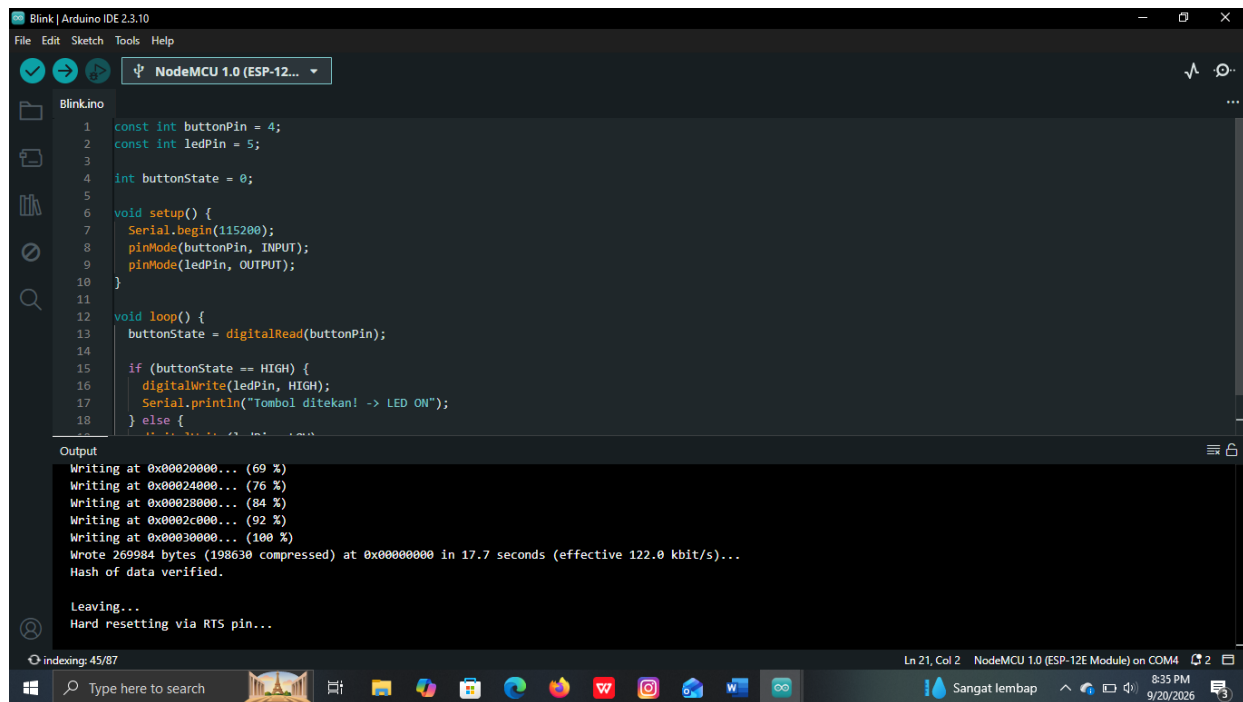
Writing at 0x00020000... (69 %)
Writing at 0x00024000... (76 %)
Writing at 0x00028000... (84 %)
Writing at 0x0002c000... (92 %)
Writing at 0x00030000... (100 %)
Wrote 270352 bytes (198837 compressed) at 0x00000000 in 17.9 seconds (effective 120.7 kbit/s)...
Hash of data verified.

Leaving...
Hard resetting via RTS pin...

Ln 16, Col 11 NodeMCU 1.0 (ESP-12E Module) on COM4 11:30 AM 9/14/2026 33°C Sebagian cerah



1.5.2 Praktikum 2: Digital Input (Membaca Push Button)



The screenshot shows the Arduino IDE 2.3.10 interface. The top menu bar includes File, Edit, Sketch, Tools, and Help. The toolbar shows icons for checking, running, and uploading. The board is set to NodeMCU 1.0 (ESP-12E Module) and the port is COM4. The sketch, named Blink.ino, contains the following code:

```
1 const int buttonPin = 4;
2 const int ledPin = 5;
3
4 int buttonState = 0;
5
6 void setup() {
7   Serial.begin(115200);
8   pinMode(buttonPin, INPUT);
9   pinMode(ledPin, OUTPUT);
10 }
11
12 void loop() {
13   buttonState = digitalRead(buttonPin);
14
15   if (buttonState == HIGH) {
16     digitalWrite(ledPin, HIGH);
17     Serial.println("Tombol ditekan! -> LED ON");
18   } else {
19     digitalWrite(ledPin, LOW);
20   }
21 }
```

The Output window shows the following messages:

```
Writing at 0x00020000... (60 %)
Writing at 0x00024000... (76 %)
Writing at 0x00028000... (84 %)
Writing at 0x0002c000... (92 %)
Writing at 0x00030000... (100 %)
Wrote 269984 bytes (198630 compressed) at 0x00000000 in 17.7 seconds (effective 122.0 kbit/s)...
Hash of data verified.

Leaving...
Hard resetting via RTS pin...
```

The status bar at the bottom indicates the current position is Ln 21, Col 2, and the board is NodeMCU 1.0 (ESP-12E Module) on COM4. The system clock shows 8:35 PM on 9/20/2026.

